Prob. 3. (2&T Ch-8. 1.1)

Show that $u(x_1, \dots x_n, t) = F(\xi_1 x_1 + \dots + \xi_n x_n - t)$, where ξ, $\xi = (\xi_1, \dots \xi_n, -1)$ is a characteristic vector, is a sol-n to the wave eq-n:

(1) $$u_{x_1 x_1} + \dots + u_{x_n x_n} - u_{tt} = 0 .$$ Fix $|\xi_t| = 1$

Sol-n.

[Preamble] Char. eq-n: $\xi_1^2 + \dots + \xi_n^2 - (\xi_t)^2 = \xi_1^2 + \dots + \xi_n^2 - 1^2 = 0$

Consider

(2) $$u(x_1, x_2, \dots x_n, t) = F(\xi_1 x_1 + \dots + \xi_n x_n - t) .$$

Partials of (2):

(3) $$\begin{cases} u_{x_1 x_1} = F''(\xi_1 x_1 + \dots + \xi_n x_n - t) \cdot \xi_1 \cdot \xi_1 \\ \dots \\ u_{x_n x_n} = F''(\xi_1 x_1 + \dots + \xi_n x_n - t) \cdot \xi_n \cdot \xi_n \\ u_{tt} = F''(\xi_1 x_1 + \dots + \xi_n x_n - t) \cdot (-1) \cdot (-1) \end{cases}$$

Putting this into (1) gives

$$F''(\xi_1 x_1 + \dots + \xi_n x_n - t) \cdot \xi_1^2 + \dots + F''(\xi_1 x_1 + \dots + \xi_n x_n - t) \cdot \xi_n^2 -$$

$$- F''(\xi_1 x_1 + \dots + \xi_n x_n - t) \cdot (-1) \cdot (-1) =$$

$$= F''(\xi_1 x_1 + \dots + \xi_n x_n - t) \cdot [\xi_1^2 + \dots + \xi_n^2 - \overbrace{(-1) \cdot (-1)}^{=1}] =$$

$$= F''(\xi_1 x_1 + \dots + \xi_n x_n - t) \cdot \underbrace{[\xi_1^2 + \dots + \xi_n^2 - 1]}_{=0 \text{ by the charact. eq-n in Preamble}} = F''(\cdot) \cdot 0 \equiv 0 \checkmark$$

Hence u satisfies the (1). ∎

$$= \frac{2}{\pi n^2}\Big[\cos(n\theta)\Big]_{\theta=0}^{\pi} = \frac{2}{\pi n^2}\cdot\Big[\cos(\pi n) - 1\Big].$$

(7) $a_n = \frac{2}{\pi n^2}\Big[\cos(\pi n) - 1\Big]$. $n = 1, 2 \ldots$

similarly

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi}\Big(|\theta| - \frac{\pi}{2}\Big)\cdot\sin(n\theta)\,d\theta = \frac{1}{\pi}\Big\{\overbrace{\int_{-\pi}^{\pi}|\theta|\cdot\sin(n\theta)\,d\theta}^{\int_{-\pi}^{\pi}\text{odd}=0} - \frac{\pi}{2}\overbrace{\int_{-\pi}^{\pi}\sin(n\theta)\,d\theta}^{\text{odd}=0}\Big\}$$

$$= \frac{1}{\pi}\cdot\Big(-\frac{\pi}{2}\Big)\cdot 0 = 0. \Rightarrow$$

(8) $b_n = 0$ $n = 1, 2 \ldots$ Then (4) becomes

$$u(r,\theta) = \sum_{n=1}^{\infty}\frac{1}{n}\cdot\frac{2[\cos(\pi n)-1]}{\pi n^2}r^n\cos(n\theta) = \sum_{n=1}^{\infty}\frac{2[\cos(\pi n) - 1]}{\pi n^3}r^n\cos(n\theta)$$

(9) $$u(r,\theta) = \sum_{n=1}^{\infty}\frac{2[\cos(\pi n) - 1]}{\pi n^3}r^n\cdot\cos(n\theta),$$

Note if $n = 2k$ $\Sigma = 0$, so let $n = 2k-1$, $k = 1, 2 \ldots$

(10) $$u(r,\theta) = \sum_{k=1}^{\infty}\frac{-4}{\pi\cdot(2k-1)^3}r^{2k-1}\cdot\cos[(2k-1)\theta],$$ is the set of all sol-ns. ▨